Interpretable Dynamic Directed Graph Convolutional Network for Multi-Relational Prediction of Missense Mutation and Drug Response

Qian Gao, Tao Xu, Xiaodi Li, Wanling Gao, Haoyuan Shi, Youhua Zhang, Jie Chen, and Zhenyu Yue

***Abstract*—Tumor heterogeneity presents a significant challenge in predicting drug responses, especially as missense mutations within the same gene can lead to varied outcomes such as drug resistance, enhanced sensitivity, or therapeutic ineffectiveness. These complex relationships highlight the need for advanced analytical approaches in oncology. Due to their powerful ability to handle heterogeneous data, graph convolutional networks (GCNs) represent a promising approach for predicting drug responses. However, simple bipartite graphs cannot accurately capture the complex relationships involved in missense mutation and drug response. Furthermore, Deep learning models for drug response are often considered "black boxes", and their interpretability remains a widely discussed issue. To address these challenges, we propose an Interpretable Dynamic Directed Graph Convolutional Network (IDDGCN) framework, which incorporates four key features: 1) the use of directed graphs to differentiate between sensitivity and resistance relationships, 2) the dynamic updating of node weights based on node-specific interactions, 3) the exploration of associations between different mutations within the same gene and drug response, and 4) the enhancement of interpretability models through the integration of a weighted mechanism that accounts for the biological significance, alongside a ground truth construction method to evaluate prediction transparency. The experimental results demonstrate that IDDGCN outperforms existing state-of-the-art models, exhibiting excellent predictive power. Both qualitative and quantitative evaluations of its interpretability further highlight its ability to explain predictions, offering a fresh perspective for precision oncology and targeted drug development.**

***Index Terms*—Tumor heterogeneity, missense mutations, directed graph convolutional networks, interpretable models.**

Received 11 July 2024; revised 18 September 2024; accepted 15 October 2024. Date of publication 18 October 2024; date of current version 6 February 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62472005, Grant 62473264, and Grant 62102004, in part by the Open Fund of National Engineering Laboratory for Big Data System Computing Technology under Grant SZUBDSC-OF2024-18, in part by the Shenzhen Science and Technology Program under Grant KCXFZ20230731094001003 and Grant JCYJ20220531102817040, and in part by the Undergraduate Innovation and Entrepreneurship Training Program Project of Anhui Agricultural University under Grant X202310364430. The open-source codes and detailed documentation of IDDGCN can be found for free at https://github.com/AhauBioinformatics/IDDGCN. *(Qian Gao and Tao Xu contributed equally to this work.) (Corresponding authors: Zhenyu Yue; Jie Chen.)*

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Digital Object Identifier 10.1109/JBHI.2024.3483316

I. Introduction

Tumor heterogeneity has always been a major challenge in the field of cancer treatment and is one of the primary determinants of treatment resistance and failure [1], [2]. Mutations at different sites within the same gene can lead to varied responses to the same drug [3]. For example, patients with Epidermal Growth Factor Receptor (EGFR) mutations such as L858R, L861Q, G719A, or L747S typically respond sensitivity to Gefitinib, whereas those with the T790M mutation may exhibit resistance [4], [5]. The variability in anti-cancer drug treatment based on missense mutations highlights the complexity of cancer treatment and presents significant challenges for accurate drug response prediction methods [6].

Artificial intelligence has shown encouraging results in anti-cancer drug response prediction by utilizing multi-omics data at the cell line level [7], [8], [9], [10], but they largely concentrate on genomic data at a broad level and seldom delve into the specific associations between missense mutations and drug responses. Understanding the impact of missense mutations on drug responses is crucial, as the extreme genomic heterogeneity within and between tumors highlights the need for precise therapeutic strategies tailored to specific genetic alterations [11], [12], [13]. With the advancement of high-throughput sequencing technologies, a vast accumulation of cancer genomic data has led to the establishment of specialized databases, such as the Cancer Genome Interpreter Cancer Biomarkers Database (CGI) [14], the Cancer Biomarker database (CIViC) [15], and the Clinical Interpretation of Variants in Cancer (JAX-CKB) [16]. However, there are inconsistencies and a lack of interperability among the various databases. To address this issue, the Meta-KB [17] database was developed to harmonize variant interpretations from multiple primary cancer mutation knowledge bases. These databases provide a foundational basis for studying the complex relationships between missense mutations and drug responses.

In recent years, Graph Neural Networks (GNNs) have demonstrated outstanding capabilities in predicting drug responses due to their ability to handle heterogeneous data [18], [19], [20]. Although no current models directly explore the relationship between missense mutations and drug, the application of GNNs in the drug response domain provides valuable insights [21], [22]. For example, MOFGCN [23] constructs heterogeneous graphs that integrate cell line similarities with drug response relationships to predict drug sensitivity. DGCL [24] employs dynamic hypergraph contrastive learning to capture global relationships between drugs and genes. However, these methods generally utilize undirected graphs and fail to distinguish between sensitivity and resistance as separate relational types, limiting deeper exploration. Moreover, these models do not dynamically adjust the interaction weights among nodes for each type of drug response relationship, which is crucial for accurately capturing the complex response patterns of a single mutation to multiple drugs. This limitation hampers their ability to precisely capture and understand the dynamic changes in drug responses, potentially affecting the accuracy of prediction outcomes. At the same time, by ignoring the heterogeneity in drug responses caused by different missense mutations within the same gene, these models have limitations in fully capturing the complexity of drug responses, which restricts their application in personalized targeted therapies.

In addition, most anti-cancer drug response prediction studies focus on improving accuracy while paying less attention to the interpretability of model predictions, making the results difficult to understand and reducing trust among researchers, doctors, and patients [25]. Current deep learning approaches are primarily based on supervised learning frameworks, they can extract high-level features but often struggle to explain the relationships between inputs and outputs [26], [27]. This limitation hinders their clinical deployment and application. The application of GNNs has significantly increased for solving real-world problems, several graph-based interpretability frameworks have been proposed, such as ExplaiNE [28], an interpretability model for link prediction, which uses counterfactual reasoning to quantify changes in prediction probabilities by perturbing embeddings. PGExplainer [29] is designed for graph classification tasks, generating subgraphs to elucidate GNN predictions by highlighting important structural features. However, these methods exhibit significant inconsistencies in the quality of explanations, and most interpretations can only provide qualitative analysis, lacking quantitative measures [30]. Constructing truthful groundtruth (GT) based on extensive domain knowledge and empirical evidence can objectively evaluate the strengths and weaknesses of interpretability models [31], thus quantitatively assessing the extent to which the explanations provided by the interpretability models match the prior knowledge. However, current interpretability models applied to drug response prediction have not utilized prior knowledge of cellular or drug mechanisms to construct GT for systematic quantitative evaluation [32], [33]. They have only conducted feature importance analysis, limiting the comprehensive understanding and validation of the effectiveness and reliability of these model interpretations in the field of drug response prediction.

To address these issues, we developed an interpretable dynamic directed graph convolutional network (IDDGCN) framework that utilizes directed heterogeneous graphs and improved interpretability models to comprehensively explore the intricate relationships between drugs and missense mutations. Our primary contributions are outlined as follows:

1. We considered the impact of missense mutations on the heterogeneity of drug responses, designing a dynamic directed graph convolutional network model that addresses these heterogeneous relationships by dynamically adjusting edge weights based on the type of relationship.
2. We explored the specific associations between mutations at different sites of a single gene and drug response, thus introducing a new dimension to the interpretable prediction methods for anticancer drug-targeted therapy.
3. We enhanced the interpretability of the model by applying two standard interpretability methods and an improved version that incorporates a weighted mechanism to account for the biological significance of different data types. Additionally, we proposed a method for constructing GT to evaluate the accuracy and reliability of the interpretability models, further improving the transparency and credibility of the predictions.
4. We conducted a consistency analysis of the explanations provided by the applied interpretable model through case studies. This approach allowed us to explore the explanatory power of the interpretable model and the basis of predictions, thereby validating the rationality of the model's predictions.

II. Materials and Methods

A. Method Overview

As depicted in Fig. 1, IDDGCN integrates mutation-drug relationships and similarity networks to construct a directed heterogeneous graph, dynamically predicting the relationships of missense mutation and drug responses within interpretable GCN. Initially, we employed the Seq2Feature [34] tool to extract key features from amino acid sequences affected by genetic mutation, which serve as the biological representation of these mutations. Drug features were then extracted by utilizing a variational autoencoder (VAE) [35] to obtain latent features

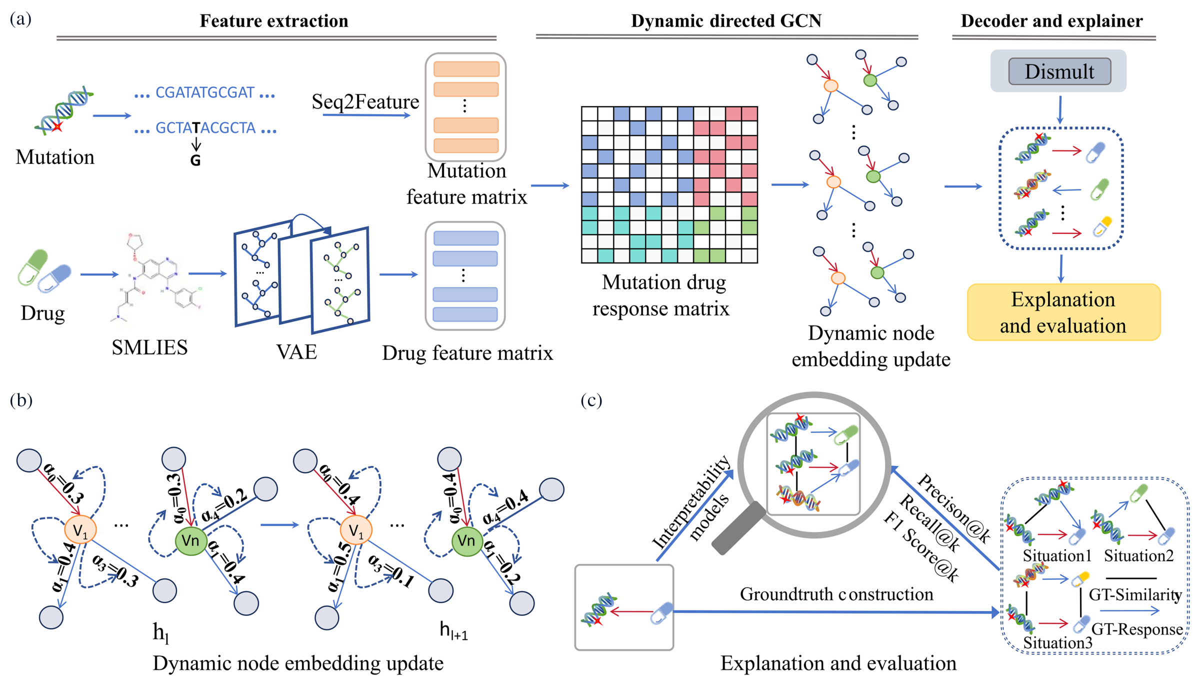

Fig. 1. Overview of the IDDGCN framework. (a) The input drug and mutation information are encoded by VAE and the feature extraction tool SeqFeature, respectively. Calculations of similarity based on these features allow us to construct networks that depict the relationships of similarity and drug responses. These relationship networks are then input into IDDGCN to learn their interactions. The predicted results are fed into interpretability models to generate explanations, which are evaluated against the constructed GT for model assessment. (b) Dynamic Node Embedding Update. V_i represents target node, the four relationship α values influencing node updates are determined by the feature embeddings of their respective nodes, thereby altering the weights of relationship edges on different nodes to explore the impact of multiple relationships in the embedding updates of different nodes. (c) Construction of GT and Evaluation of Interpretability Models Based on Drug Response and Similarity Networks. The explanations are generated using two distinct interpretability frameworks, along with an enhanced version that integrates biological information. GT is constructed using three different scenarios based on the edges to be interpreted, and the explanations are assessed using Precision@k, Recall@k, and F1 Score@k.

from SMILES strings. Based on these features, we use cosine similarity to calculate both drug and mutation similarity matrices. Subsequently, IDDGCN merges the multiple response relationships between mutations and drugs to construct a directed bipartite graph, and the inclusion of similarity networks provides additional contextual clues to improve the learning efficiency. In this heterogeneous graph, nodes represent either mutations or drugs, and edges are encoded based on four distinct attributes: sensitivity, resistance, mutation similarity, and drug similarity. In the node representation learning phase, IDDGCN dynamically adjusts the influence weights of the four relationships for updating node embeddings, further capturing the subtle variations in drug responses. Predictions are made using the updated embeddings and a DistMult decoder [36], providing insights into the complex interplay between mutations and drug efficacy. In the final stage, we explained the model's predictions using two distinct interpretability frameworks, and proposed an enhanced version which integrates a weighted mechanism to account for the biological significance of different data types, further refined the interpretability. To assess the validity of these interpretations, we constructed GT to evaluate validity of the interpretations.

B. Materials

This study principally involves the MetaKB database, the Ensembl platform [37], and a tool for extracting amino acid sequence features named Seq2Feature. The MetaKB [17] database aggregates 19,551 entries from six clinical databases, each detailing the type of gene, the amino acid mutation (e.g., E242C), and the associated cell drug response. Specifically, we have selected the missense mutation drug response data recorded in the MetaKB database as the focus of our research. Then we turned to the Ensembl database to obtain the amino acid sequences of these selected proteins. Here we utilized the GRCh37 human genome version provided by Ensembl, which allowed us to accurately retrieve and analyze the genetic information pertinent to this research.

We employed Seq2Feature, a comprehensive tool designed to extract a wide range of features from both protein and DNA sequences, to obtain mutation features from the sequences. By inputting the amino acid sequences and specifying the mutations with their positions, we were able to generate detailed features, including the biochemical properties of amino acids, substitution matrix features, and pairwise properties such as contact potential that indicate spatial proximity and interaction tendencies between amino acid residues. For the drug feature extraction part, we sourced the PubChem database to obtain the SMILES strings of drugs, which were used for subsequent feature conversion into drug characteristic matrices.

Following data preprocessing, we acquired a dataset comprising 1,754 reaction entries, drawn from 661 mutation instances and 184 drugs. These mutations are distributed across 71 distinct genes, with the most frequent being mutations in ABL1 (104 unique mutations), ERBB2 (47 unique mutations), and BRAF

(44 unique mutations). Notably, mutations in the ABL1 gene appear most frequently in the dataset, with 134 occurrences related to drug responses. Finally, this dataset includes 919 instances of resistance and 835 instances of sensitivity.

C. Multi-Relational Prediction of Missense Mutation and Drug Response

1) Cell Similarity and Drug Similarity: Based on the view that similar genetic mutations exhibit similar drug sensitivity [38], [39], our study conducted analyses of mutation similarity and drug similarity to infer the sensitivity of anti-cancer drugs. Mutation similarity was quantified using features extracted from amino acid sequences altered by genetic mutation. These features include but are not limited to biochemical properties and spatial configurations which are critical for understanding the functional impacts of mutations. To better assess drug similarity, we employed a VAE [35] to extract features from the SMILES strings representing the molecular structures of drugs. These methods embed molecular structures into a continuous latent space that captures the underlying structural and functional characteristics of the drugs. Both mutation similarity S_m and drug similarity S_d are calculated using the cosine similarity, defined as:

$$S(i,j) = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|} = \frac{\sum_{k=1}^{n} v_{ik} \cdot v_{jk}}{\sqrt{\sum_{k=1}^{n} v_{ik}^2} \sqrt{\sum_{k=1}^{n} v_{jk}^2}} \tag{1}$$

Here, $V_i$and V_j represent the feature vectors for entities i and j respectively, with n indicating the dimensionality of these vectors. The calculation utilizes the product of the Euclidean norms of the vectors to normalize the result, ensuring that the values range between -1 and 1, thereby capturing the degree of alignment between the feature vectors. We have set stringent thresholds for similarity acceptance to ensure the robustness of similarity networks: a cosine similarity score of 0.97 is required for considering two mutations as similar, while a score of 0.78 is necessary to establish drug similarity. These thresholds were determined through experimental analysis, in which we evaluated the proportion of similarity data relative to the total dataset and its impact on predictive performance. By systematically varying the similarity thresholds and calculating the area under the precision-recall curve (AUPR) for each configuration, we observed that when the similarity data represented around 3% of the total pairs in the mutation matrix and drug matrix, (with thresholds of 0.97 for mutations and 0.78 for drugs), the AUPR for the prediction task was maximized. This indicates that the selected thresholds provided the optimal balance between similarity data and the accuracy of mutation-drug response predictions.

2) Dynamic Directed Graph Convolutional Network: In this study, we utilize Directed Graph Convolutional Neural Networks, an approach that integrates node attributes, network structure, and edge directionality to derive representative vector embeddings [40]. Firstly, we constructed a directed heterogeneous graph $\mathcal{G} = (V, \mathcal{E}, \mathcal{R})$ to intricately describe the complex interactions between drugs and missense mutations. The node set V comprises drug nodes V_d and mutation nodes V_m, with the feature matrix $X \in \mathbb{R}^{N \times F}$ where N denotes the total number of nodes and F represents the dimensionality of feature embeddings. The initial node embeddings were randomly initialized from a uniform distribution ranging from 0 to 1. Preliminary experiments showed this method outperformed traditional biological feature-based initialization, revealing unexpected advantages for our tasks. The edge set includes edges labeled with types of relationships $(s, r, o) \in \mathcal{E}$, where s represents the subject (e.g., a mutation node V_m), o represents the object (e.g., a drug node V_d), and $r \in \mathcal{R}$ indicates the type of relation. The set of relationship types $\mathcal{R} \in \{\mathcal{R}_0, \mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3\}$ consists of four main relations. Specifically, we designed directed edges to capture different types of interactions: edges from drugs to cell lines represent resistance $\mathcal{R}_0$, while edges from cell lines to drugs represent sensitivity $\mathcal{R}_1$. This bidirectional design allows us to encode the directional nature of drug response relationships within the graph. Additionally, we incorporated drug similarity $\mathcal{R}_2$ and mutation similarity $\mathcal{R}_3$ into the graph structure to enrich the information and supplement the data necessary for node prediction.

Considering the heterogeneity of the bipartite graph representing the associations between mutations and drug responses, inspired by refs [41], we have incorporated a dynamic weighting mechanism into the original Relational Graph Convolutional Network (RGCN) node feature update process. This mechanism allows the model to dynamically adjust the weights of different types of relationships for node embedding updates based on node features, enabling more precise capture and reflection of these relationships in predicting drug responses. The formula is as follows:

$$h_i^{(l+1)} = \sigma\left(\sum_{r \in R} \alpha_r^{(l)} \sum_{j \in N_i^r} \frac{1}{c_{i,r}} W_r^{(l)} h_j^{(l)} + W_0^{(l)} h_i^{(l)}\right) \tag{2}$$

Where $h_i^{(l)}$ represents the embedding of node V at the $l-th$ layer, N_i^r designates the set of neighbor indices associated with node V through relation r, $W_r^{(l)}$ is the relation-specific weight matrix distinguishing different relations, σ is the sigmoid function that maps model outputs to the [0, 1] interval, and $\alpha_r^{(l)}$ is the added dynamic weight parameter, calculated as:

$$\alpha_r^{(l)} = \text{softmax}\left(W_\alpha^{(l)} h_i^{(l)} + b_\alpha^{(l)}\right) \tag{3}$$

Here, $W_\alpha^{(l)}$ and $b_\alpha^{(l)}$ are trainable parameters used to adjust the weight of each relation based on the features $h_i^{(l)}$ of node V at layer l.

3) DistMult Decoder: After learning the final embeddings of mutations and drugs, we employed DistMult [36] as the scoring function to assess the validity of triples (s, r, o). DistMult calculates the score of triples through the following equation:

$$f(s, r, o) = h_s^{(f)T} D_r h_0^{(f)} \tag{4}$$

Where $h_s^{(f)}$ and $h_0^{(f)}$ represent the embedding vectors of the head and tail entities respectively, and D_r is the diagonal matrix representation of the relation r.

To effectively train this model, we generate negative samples by randomly altering the head or tail entities of known positive samples [42]. We chose a 1:1 ratio of positive to negative samples for balance, this ratio is adjustable to accommodate different training needs and data characteristics. This approach helps us assign corresponding negative samples to each positive sample, allowing the model to better differentiate relationships. Then we use the cross-entropy loss function to enhance model performance, which considers the predictive probabilities of positive and negative samples:

$$\text{loss} = -\frac{1}{|\beta|} \sum_{(s,r,o,y)\in\beta} \Big[y \log(\sigma(f(s,r,o))) + (1-y)\log(1-\sigma(f(s,r,o))) \Big] \tag{5}$$

Here, β represents the total of positive and negative samples in the training set, and y is the label. Introducing negative samples and using this loss function enable the model to simultaneously learn to attract positive samples and repel negative ones during training, ensuring that it fully leverages the complex interactions between mutations and drugs.

D. Interpretability of IDDGCN Predictions

1) Explainable Predictions With Graph Neural Networks: We applied two advanced explainability frameworks, ExplaiNE [28] and GNNExplainer [43], to identify and quantify the contributions of specific triples to the prediction results. Additionally, we further developed an enhanced model that preserves the biological integrity of the explanatory data structure during the interpretation process. By inputting the trained node embeddings and the edges requiring explanation into these frameworks, we leverage their distinct mechanisms to calculate the contribution scores of neighboring edges to the target edge. These approaches provide a robust metric to assess the influence of each edge on the prediction outcomes, thereby offering a more nuanced insight into the model's decision-making process.

ExplaiNE utilizes counterfactual reasoning to explore the impact of specific links within the network structure. By quantifying the importance of links through derivatives, it effectively assesses how alterations in network connections influence prediction outcomes, thus providing critical insights into the model's decision-making process.The computation is given by the following formula:

$$\frac{\partial p(s,r,o)}{\partial a_{kl}}(A) = \nabla_X p(s,r,o)(H^*)^T \cdot \frac{\partial H^*}{\partial a_{kl}}(A) \tag{6}$$

Here, $p(s,r,o)$represents the probability of an relationship r existing between nodes s and o. The indices k and l refer to candidate nodes selected from the adjacency matrix A of the edge, H^* denotes the final node embeddings obtained through training the predictive model.

The objective of GNNExplainer is to maximize the mutual information between the model's predictions on the original graph and the subgraph after applying a mask. By optimizing the mask matrix, it aims to increase the mutual information between the original predictions and those after masking, ensuring that the masked subgraph retains the structural features most influential to the predictions. The minimization objective function is as follows:

$$P_{loss} = \min_M - \sum_{i=1}^{R} \mathbb{1}[r=i] \log P_\Phi(r'|A_i \odot \sigma(M), H^*) \tag{7}$$

Here, M is the learned mask matrix, which is used to identify the most impactful parts of the graph for the model's predictions. A_i represents the adjacency matrix for category H^* is the feature matrix used in the prediction model, and $[r=i]$ is an indicator function that is 1 when the actual label r equals category i.

Building upon the GNNExplainer, we provided a weighting mechanism and refined the loss function to accommodate the specific characteristics of biological data. In biological datasets, different data types often vary significantly in their importance. To account for this, we introduced a new loss function with a weighting mechanism to address this issue. We consider (7) as the prediction loss P_{loss}, and further introduce a biological structure loss B_{loss} in the loss function calculation as follow:

$$B_{\text{loss}} = \frac{1}{r}\sum_{i=1}^{r}(T_i - C_i)^2 \tag{8}$$

Here, T_i represents the target ratio, allowing that the adjacency matrix generated by the mask for a specific edge retains the desired proportions throughout optimization. Given that our dataset contains a higher proportion of similarity data and relatively fewer sensitivity data points, we set T_i to $[0.4, 0.4, 0.1, 0.1]$ to maintain that the explanatory graph focuses more on the predominant data features. C_i denotes the actual ratio after masking, and r is the total number of relations, used to compute the mean error across all relations. The overall loss function is expressed as:

$$LOSS_{all} = \frac{B_{\text{loss}} + S_{\text{loss}}}{2} \tag{9}$$

In this formulation, P_{loss} accounts for the loss based on prediction outcomes, measuring the impact of the masked graph structure on the model's predictions, while B_{loss} ensures that the explanations for a specific edge remain consistent with the target ratios throughout the process of generating explanations. This dual-focus loss function design emphasizes preserving the biological relevance of the explanations, making them more reliable and consistent with the underlying properties of the data.

2) Interpretability Assessment With Established Groundtruth: To evaluate the explainability of link predictions in missense mutation-drug response network, we provided a method for constructing a dedicated GT dataset within a predictive GCNs. This set encompassed multiple verifiable explanations for each drug-mutation pair, providing a comprehensive benchmark for evaluation. To quantify the performance of the explainability methods, we introduced a series of metrics to establish an extensive evaluation framework aimed at revealing the strengths and limitations of the model's explainability.

GT were constructed based on existing similarity and response relationships between mutations and drugs. As shown in

Fig. 1(a), this set was primarily categorized into three situations for a specific drug-mutation pair (m_1, d_1) to be explained.

Situation 1: Based on mutation similarity, if there exists m_2 similar to m_1 that has the same response relationship with d_1, then (m_1, m_2) serves as a GT-Similarity, and (m_2, d_1) as a GT-Response.

Situation 2: Based on drug similarity, if there exists d_2 similar to d_1 that has the same response relationship with m_1, then (d_1, d_2) serves as a GT-Similarity, and (m_1, d_2) as a GT-Response.

Situation 3: Based on both mutation and drug similarities to expand the scope of the groundtruth, if there exists a mutation m_2 similar to m_1, a drug d_2 similar to d_1 and (m_2, d_2) has the same response relationship with (m_2, d_2), then (m_1, m_2), (d_1, d_2) serves as a GT-Similarity, and (m_2, d_2) the GT-Response.

To assess the accuracy of the interpretability models, we compare the contribution with the GT using three metrics: Precision@k, Recall@k and F1@k, calculated using the formula:

$$\text{Precision@k} = \frac{\text{true positives@k}}{k} \tag{10}$$

$$\text{Recall@k} = \frac{\text{true positives@k}}{\text{TG}} \tag{11}$$

$$\text{F1@k} = \frac{2 * \text{Precision@k} * \text{Recall@k}}{\text{Precision@k} + \text{Recall@k}} \tag{12}$$

$Precision@k$ represents the proportion of true positives among the top k explanations provided by the model, k represents the top k explanations ranked by their contribution as provided by the model, true positives@k denotes the number of correct true positive explanations among the top k provided by the model. $Recall@k$ measures how many of the true positives within the GT are correctly explained by the model within the top k predictions, TG represents the number of overlaps between the explanations provided by the model and GT. $F1@k$ provided a balanced measure of precision and recall, these metrics reflect both the accuracy of the model and its coverage of real situations, serving as crucial standards for evaluating the performance of interpretability models.

We use the PyTorch framework to implement the model codes and the hyperparameters are tuned by grid searching. Here, we set the batch size to 100, the learning rate to 0.001, and the embedding dimensionality to 64. We used a three-layer GCN architecture, and the number of neural network iterations was 5000 epochs.

III. Result

A. Experimental Design

To validate the effectiveness of our model, we compared it with two machine learning models, four deep learning models. The models included random forest (RF), XGBOOST, MOFGCN [23], NIHGCN [44], DGCL [24] and RGCN [45]. MOFGCN constructs a heterogeneous graph by integrating cell line similarity, drug similarity, and cell line-drug responses to explore potential relationships between cell lines and drugs. NIHGCN, an innovative heterogeneous graph convolution network model, precisely predicts anticancer drug responses through parallel graph convolution layers and neighborhood interaction layers. The DGCL framework enhances data representation through dynamic hypergraph structures and innovatively uses contrastive learning to address common issues of data sparsity and over-smoothing in graph neural networks, specifically targeting the prediction of multi-relational drug-gene interactions. RGCN as a GCN model for handling relational networks, was applied to our directed heterogeneous network constructed from mutation drug response data and similarity networks to compare with our improved IDDGCN. All algorithms used for comparison were configured with the hyperparameters recommended in their respective original papers. We tested IDDGCN and all baseline methods on the same dataset, and performed comparisons using ACC, Precision, Recall, F1, AUROC, and AUPRC as evaluation metrics [46].

B. Comparison of Model Performance in Five-Fold Cross-Validation

We prepared an equal number of negative samples for both resistance and sensitivity relationships, representing either no relationship or undiscovered connections, achieving a balanced dataset. To validate the performance of IDDGCN in the multi-relational prediction of missense mutation and drug response, we conducted a five-fold cross-validation. The data was then divided into five equal parts, and in each round of validation, 1/5 of the samples from the association matrix were selected as the test set, while the remaining 4/5 were used as training data. This approach ensured the absolute separation of training and testing datasets, maintaining the fairness of our model evaluation.

Table I presents the results of machine learning models, state-of-the-art models, RGCN, and IDDGCN under five-fold cross-validation, where IDDGCN outperformed existing models on various metrics in predicting drug responses. Furthermore, as illustrated in Fig. 2(a) and (b), both the Receiver Operating Characteristic (ROC) curve (Fig. 2(a) and the Precision-Recall (PR) curve (Fig. 2(b) highlight IDDGCN's effectiveness, achieving an AUROC of 0.8994 and an AUPRC of 0.9366, respectively. These results demonstrate that IDDGCN not only delivers superior overall performance but also maintains high precision across different recall thresholds. In clinical settings, the accuracy of drug response predictions is critical, as it can have a significant impact on patient treatment outcomes. Additionally, the RGCN model also showed robust performance, achieving the second-highest scores in most categories, except accuracy. With AUROC and AUPRC values of 0.8813 and 0.9213, this approach indicates that incorporating sensitivity and resistance into a directed graph as two distinct relationships for deep exploration is highly significant for predicting correlations between missense mutations and drug responses. The observation that all metrics for RGCN fall below those of IDDGCN further demonstrates the advantage of dynamically adjusting edge weights based on different response relationships in enhancing predictive accuracy.

TABLE I
COMPARISON OF EACH METHOD UNDER FIVE-FOLD CROSS-VALIDATION

Method	Accuracy	Precision	Recall	F1 Score	AUROC	AUPRC
RF	$0.8143{\pm}3{\times}10^{-3}$	$0.6736{\pm}9{\times}10^{-3}$	$0.4750{\pm}1{\times}10^{-2}$	$0.5566{\pm}1{\times}10^{-2}$	$0.8301{\pm}6{\times}10^{-3}$	$0.6250{\pm}7{\times}10^{-3}$
XGBOOST	$0.8138{\pm}5{\times}10^{-3}$	$0.5664{\pm}1{\times}10^{-3}$	$0.4953{\pm}1{\times}10^{-3}$	$0.6620{\pm}1{\times}10^{-2}$	$0.8323{\pm}3{\times}10^{-3}$	$0.6444{\pm}1{\times}10^{-2}$
MOFGCN	$0.7425{\pm}1{\times}10^{-2}$	$0.7096{\pm}2{\times}10^{-2}$	$0.8323{\pm}2{\times}10^{-2}$	$0.7642{\pm}5{\times}10^{-3}$	$0.7978{\pm}7{\times}10^{-3}$	$0.8119{\pm}8{\times}10^{-3}$
NIHGCN	$0.7814{\pm}2{\times}10^{-2}$	$0.8589{\pm}2{\times}10^{-2}$	$0.6743{\pm}2{\times}10^{-2}$	$0.7552{\pm}2{\times}10^{-2}$	$0.8595{\pm}2{\times}10^{-3}$	$0.8624{\pm}1{\times}10^{-2}$
DGCL	$0.8330{\pm}4{\times}10^{-3}$	$0.8062{\pm}1{\times}10^{-2}$	$0.6822{\pm}1{\times}10^{-3}$	$0.7384{\pm}5{\times}10^{-3}$	$0.8752{\pm}1{\times}10^{-3}$	$0.7984{\pm}5{\times}10^{-3}$
RGCN	$0.8280{\pm}7{\times}10^{-3}$	$0.8713{\pm}4{\times}10^{-3}$	$0.8572{\pm}9{\times}10^{-3}$	$0.8641{\pm}6{\times}10^{-3}$	$0.8813{\pm}1{\times}10^{-3}$	$0.9283{\pm}4{\times}10^{-3}$
IDDGCN	$\mathbf{0.8440{\pm}5{\times}10^{-3}}$	$\mathbf{0.8866{\pm}6{\times}10^{-3}}$	$\mathbf{0.8786{\pm}5{\times}10^{-3}}$	$\mathbf{0.8825{\pm}4{\times}10^{-3}}$	$\mathbf{0.8994{\pm}6{\times}10^{-3}}$	$\mathbf{0.9366{\pm}5{\times}10^{-3}}$

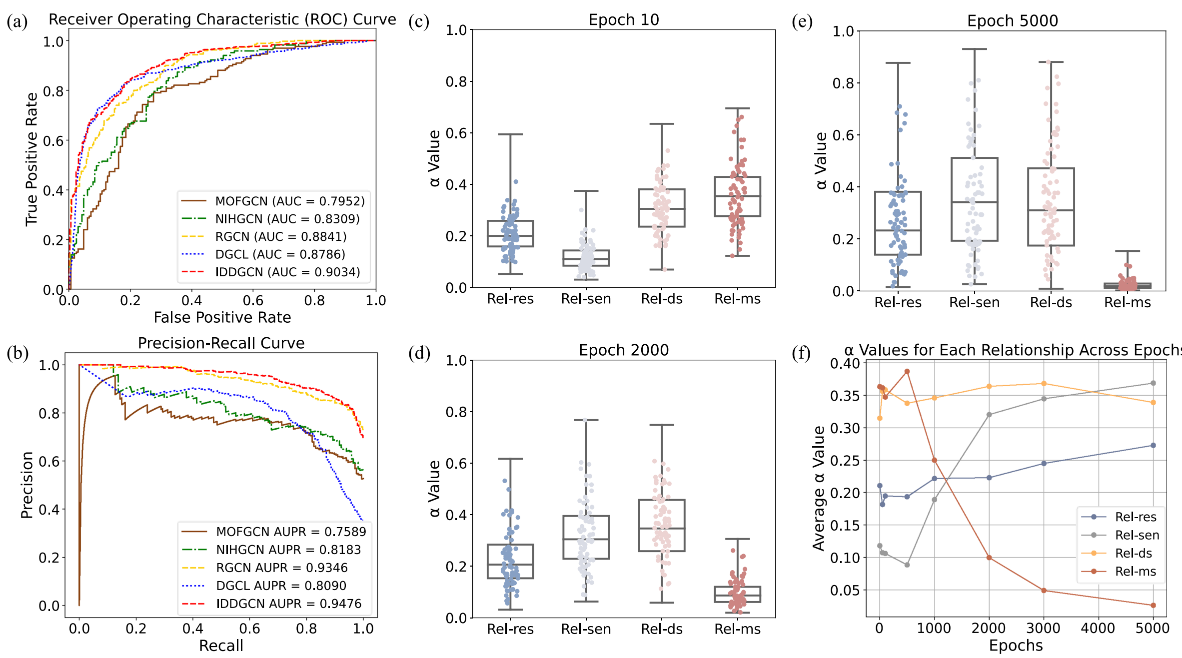

Fig. 2. ROC and P-R Curves, and Analysis of α Values Across Epochs Under IDDGCN and Baselines on Independent Test. (a), (b) ROC and P-R curves obtained under IDDGCN and baseline on the independent test. (c)–(e) α values for four different relationships at different epochs under IDDGCN. (f) Changes in average α values for each relationship across different epochs under IDDGCN. Rel-res, Relation resistance; Rel-sen, Relation sensitivity; Rel-ds, Relation drug similarity; Rel-ms, Relation mutation similarity.

C. Dynamic Adjustment of Impact Factors in Embedded Updates

We analyzed the α values when updating node embeddings to quantify the influence of different relationship types on the updating of node features. As illustrated in Fig. 2(c), (d), and (e), the distribution of α values for the four types of relationships at different epochs reveals the dynamic adjustment process within the IDDGCN model. At epoch 10 (Fig. 2(c)), the α values for Relation resistance (Rel-res) and Relation sensitivity (Rel-sen) are relatively low and concentrated, whereas Relation drug similarity (Rel-ds) and Relation mutation similarity (Rel-ms) exhibit higher and more dispersed values. By epoch 2000 (Fig. 2(d)), the α values for Rel-res and Rel-sen have started to increase, while those for Rel-ds and Rel-ms show a downward trend. This adjustment becomes more pronounced at epoch 5000 (Fig. 2(e)), where Rel-res and Rel-sen significantly rise in α values, indicating their increasing importance, while Rel-ds and Rel-ms continue to decline. Fig. 2(f) further corroborates this trend by depicting the changes in average α values for each relationship type across epochs. It shows that the average α values for Rel-res and Rel-sen progressively increase throughout the training process, whereas those for Rel-ds and Rel-ms decrease. This adaptability in adjusting edge weights based on response relationship types underscores the efficacy of the IDDGCN model in capturing the complex interactions within heterogeneous networks.

D. Analysis of IDDGCN in Gene-Specific Test Sets

To evaluate the generalization performance of IDDGCN in distinguishing the specific associations between mutations at different sites of a single gene and drug response, we selected drug response data for two genes that are more frequently represented and well-studied (ABL1 and KRAS) [47], [48]. These experiments are crucial for exploring the model's ability to identify subtle differences in mutations within one gene and their association with drug response.

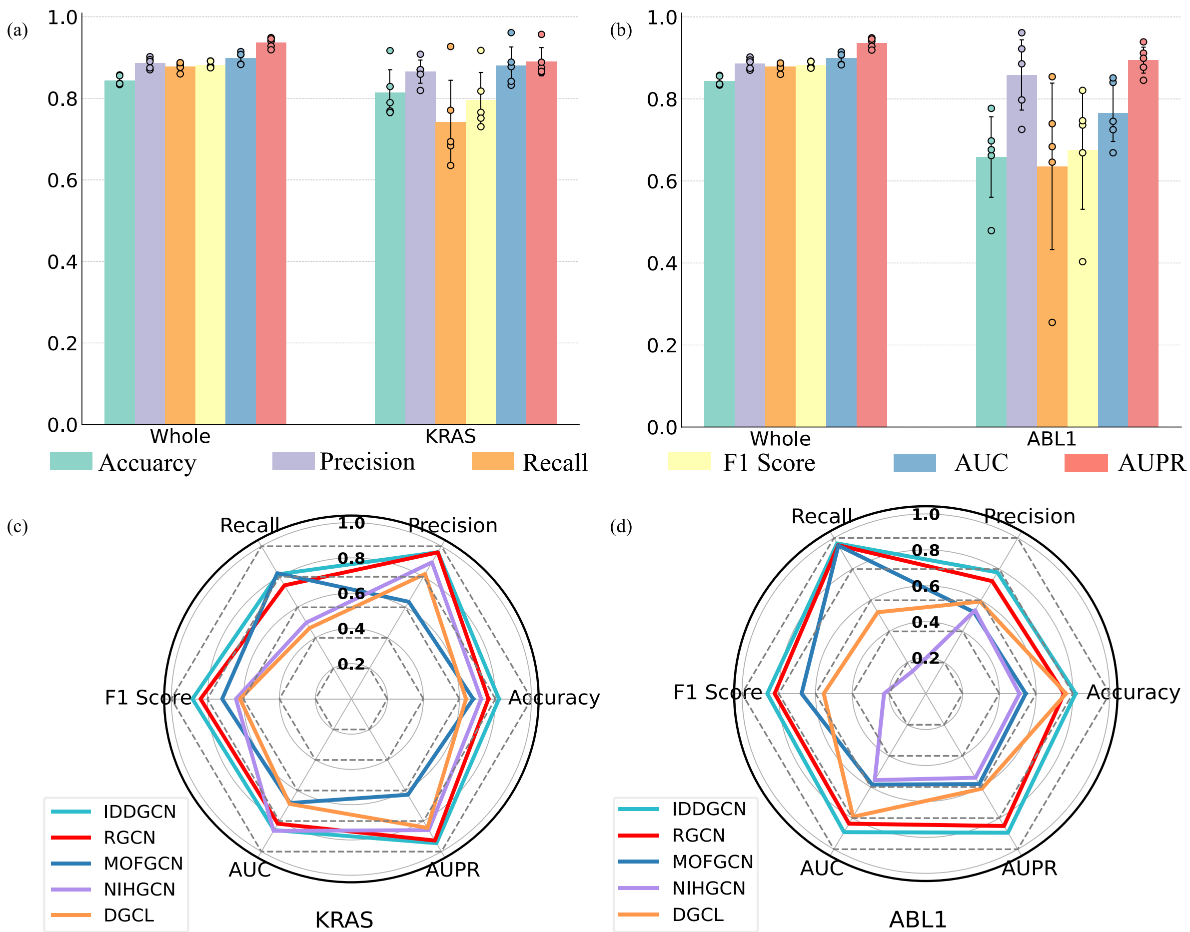

Fig. 3. Performance comparison of IDDGCN with other methods in the intragenic experiments and the independent test set experiments. (a) Performance comparison for the ABL1 and the whole sets. (b) Performance comparison for the KRAS and the whole sets. (c) Comparison of various model prediction metrics in the ABL1 test set. (d) Comparison of various model prediction metrics in the KRAS test set.

1) Intragenic Experiments Evaluation for ABL1 and KRAS: Initially, we focused on intragenic experiments for the ABL1 and KRAS. For the ABL1, 104 mutations and 10 drugs were selected, resulting in 217 response edges. For the KRAS, we selected 34 mutations and 48 drugs, resulting in 242 response edges. Then we reconstructed mutation similarity and drug similarity networks and compared the results with those obtained from the whole dataset. As shown in Fig. I(a) for ABL1 and Fig. I(b) for KRAS, the performance on the single gene datasets was slightly lower than that on the whole datasets, but the metrics still demonstrated solid predictive capabilities. Despite the smaller dataset, the IDDGCN model still can effectively capture the critical relationships between mutations and drug responses within a single gene, showing its potential in biomarker development.

2) Independent Test Set Evaluation for ABL1 and KRAS: We used all mutation-drug response data involving the ABL1 or KRAS in the dataset as independent test sets. To ensure data purity, we removed the related similarity edges involving these genes in both the IDDGCN and RGCN models, maintaining consistency across other comparison models. For the ABL1, 1,537 response edges involving 557 mutations were obtained as the training set and 217 response edges involving 104 missense mutations as the test set. For the KRAS, we obtained 1,512 response edges involving 627 mutations as the training set and 242 response edges involving 34 mutations as the test set. The model was compared with several advanced models and RGCN using metrics such as accuracy, precision, recall, F1 score, AUROC, and AUPRC. As shown in Fig. 3(c) for ABL1 and Fig. 3(d) for KRAS, our model outperformed other models across all metrics. Specifically, IDDGCN exhibited superior accuracy, precision, and recall, indicating its effectiveness in predicting drug sensitivity and resistance to mutations.

Through these experiments, we demonstrated that IDDGCN exhibits high robustness and reliability in handling a new gene test set and distinguishing the subtle differences in drug responses caused by different mutations within a gene. This capability is crucial for understanding the complexity of drug responses and provides significant support for targeted drug development. But it was observed that the metrics for ABL1 were consistently lower than those for KRAS in both the intragenic experiments and the independent test set experiments. This discrepancy may be attributed to the higher number of missense mutations in ABL1, leading to a sparser dataset and making it more challenging in handling complex intragenic relationships.

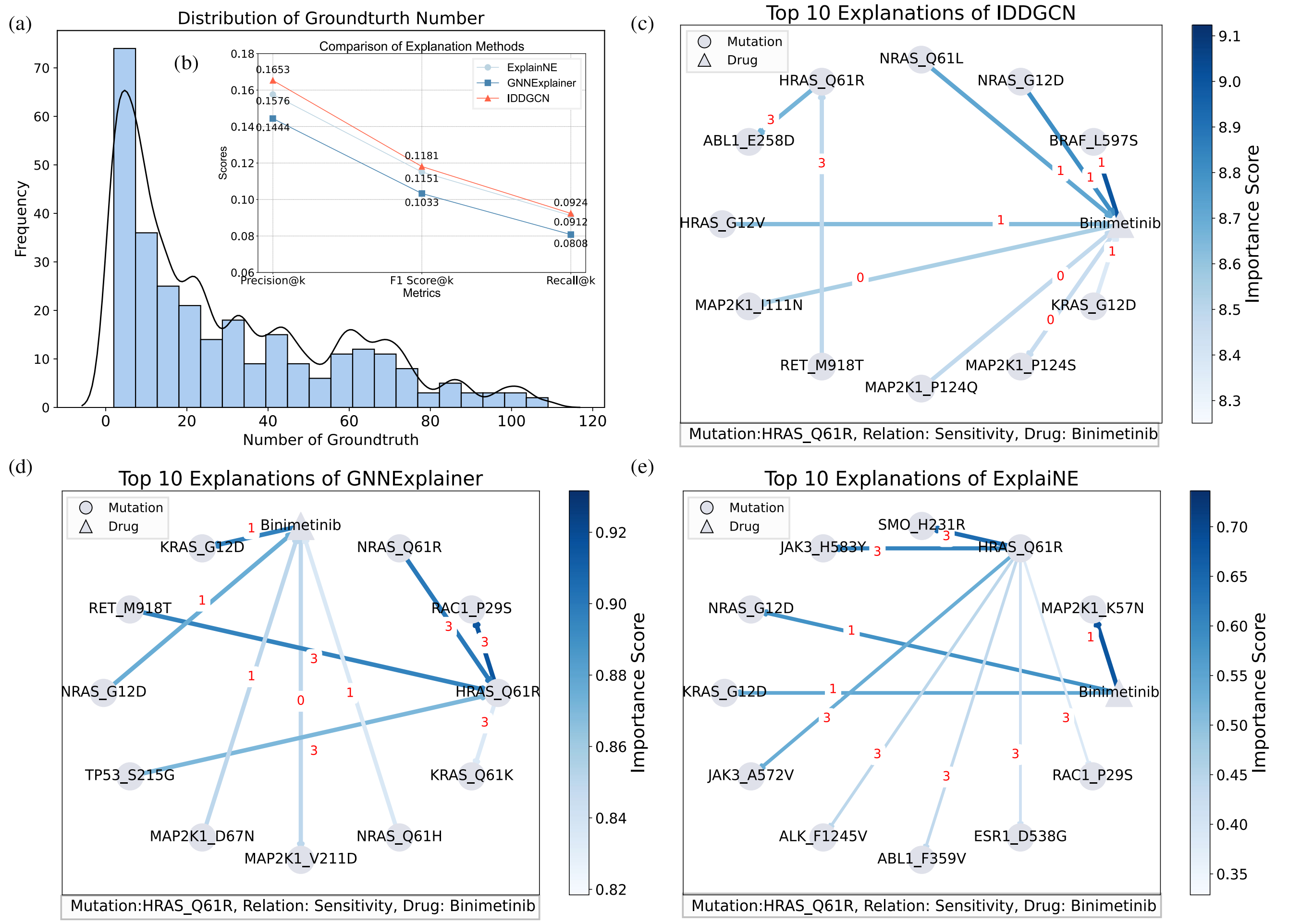

Fig. 4. Analysis and Evaluation of Interpretability Models. (a) Distribution of the constructed GT. (b) Evaluation of interpretability models based on GT. (c) Visualization of top 10 explanations provided by IDDGCN for the HRAS_Q61R and Binimetinib pair. (d) Visualization of top 10 explanations provided by GNNExplainer for the HRAS_Q61R and Binimetinib pair. (e) Visualization of top 10 explanations provided by ExplaiNE for the HRAS_Q61R and Binimetinib pair. Relation resistance:0; Relation sensitivity: 1, Relation drug similarity:2, Relation mutation similarity:3.

E. Analysis of Explanation on Predictions in IDDGCN

1) Construction of Groundtruth: Following the methods described in the "Methods" section, three types of GT were generated to evaluate the explanation of the prediction model. As shown in Fig. 4(a). Most GT ranged between 5 and 20 edges, with the maximum being 109 edges, and a few cases having no GT at all. The variability in the distribution reflects the complexity of biological interactions, while the absence of GT edges in certain cases suggests that the construction method may need further refinement, or these areas lack sufficient biological data support. For model evaluation, we retained only the edges with 3 or more GT, as edges with fewer than 3 GT may not provide sufficient data for accurate assessment.

2) Evaluation of Interpretability Models: We use Accuracy@k, Precision@k, and Recall@k to evaluate the alignment between the interpretability models and the GT. We set k to 5, meaning that the top 5 explanations provided by the interpretability model were compared with the GT, the test results are presented in Fig. 4(b). IDDGCN outperforms both ExplaiNE and GNNExplainer across all three metrics, indicating that the weighted mechanism significantly enhances alignment with the GT by better capturing the biological significance of different data types, leading to more accurate and comprehensive explanations of the prediction results.Despite these improvements, the overall metrics for both interpretability models are relatively low, this may due to the complexity of the data and model, the incompleteness of the GT construction method, and the strict evaluation criteria.

3) Visualization of Top Contributing Edges: We performed a visualization analysis of the results of the interpretability models, displaying the top ten edges contributing to the prediction results, as shown in Fig. 4(c), (d), and (e). Due to the different contribution calculation methods of these models, there are some discrepancies in the metrics. The charts show the top ten important explanatory edges for the sensitivity prediction between the gene mutation HRAS_Q61R and the drug Binimetinib in both models [49]. It can be observed that there are two overlapping edges appear in all three models: the sensitivity relationships the sensitivity relationships NRAS_G12D-Binimetinib, KRAS_G12D-Binimetinib. Notably, NRAS and KRAS, similar to HRAS, are members of the RAS gene family, and both NRAS and KRAS mutations involve the substitution of glycine with aspartic acid at position 12 (G12D). These mutations result in the constitutive activation of the MAPK/ERK signaling pathway, which drives cellular proliferation and survival [50]. The identification of these overlapping edges demonstrates the basis provided by the interpretability models for IDDGCN predictions, further strengthening the biological plausibility and robustness of our predictive model. Additionally, the IDDGCN identified that edges associated with the RAS family or MAP2K1 play a dominant role in the explanations for the HRAS_Q61R and Binimetinib pair, which underscores the model's sensitivity to key oncogenic pathways involved in drug response. The presence of these shared and biologically relevant edges highlights the importance of key pathways and mutations in determining drug sensitivity and emphasizes how the integration of multiple interpretability approaches strengthen the validation and reliability of model predictions.

IV. Conclusion

We proposed IDDGCN as a comprehensive framework for predicting the relationships of missense mutation and drug responses and providing explanations and evaluations for these predictions. First, IDDGCN integrates resistance and sensitivity as two relationships into the directed graph, combining mutation and drug similarity information to construct a directed heterogeneous network to explore complex response relationships. Second, to further explore the associations between different mutations and drug responses, we dynamically adjust the weights of various relationship types based on their influence on the updating of node features. Third, we enhanced the model's interpretability by integrating it with two established link prediction interpretability algorithms and a modified version that incorporates a weighted mechanism to account for the biological significance of different data types. This combination allows for a more nuanced exploration of how our model processes and predicts data. Last, we constructed GT based on the characteristics of drug responses to evaluate whether the explanations provided by the interpretability algorithms align with human understanding.

However, there are still areas for improvement. We have established a comprehensive framework that addresses both explainability and prediction, but these two aspects are not yet integrated to enhance the prediction outcomes. Furthermore, the approach for constructing GT, which uses similarity and response relationships, is relatively simple. In the future, we will improve the interpretability models to provide more detailed and biologically meaningful explanations. This includes incorporating more biological knowledge and rules to enhance the credibility and operability of the interpretations. We will also explore how to integrate different interpretability methods to provide comprehensive, multi-perspective explanations. Additionally, we will introduce more dimensions of features and relationships, such as gene function, signaling pathways, and cellular environments, to construct a more complex and accurate GT. This will help us to more comprehensively evaluate the performance of interpretability models and enhance their credibility in practical applications. Finally, we will explore how to use interpretability results to improve the prediction model. By analyzing interpretability results, we can identify weak points and potential biases in the prediction model and make targeted optimizations.

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